



# Automated Palo Alto firewall policies conversion to Versa service template

Author: Loi Ngo, loi@versa-networks.com

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Update 1.1: I fixed some minor oversights in the policy conversion script.

Update 1.2: I added support for PAN native XML format config files. Also added automated address object creation and replacement when policy contains an IP address or subnet.

**Purpose:** As the title of this document says, this document describes what the set of scripts do and can't do. The scripts is capable of cleaning, extracting, modifying and converting Palo Alto firewall policies and its dependent objects to Versa configuration in the form of service template. The whole process is fully automated, except when there are unresolved objects and/or configuration that must manually address. In such case, you'll be given the chance address the issues before the scripts do the conversion.

## What is supported:

- PAN policies that use ip-address objects, nested objects groups, predefined/custom services, predefined applications, custom URL categories, security URLF profiles, default policies security profiles and LDAP DN users/groups.
- Different types of objects: static ip-address, ip-range or fqdn-objects
- Recognize ip-groups that are being used in policies
- Import custom services objects, recognize which objects are being used in policies
- Convert PAN predefined services references to Versa predefined services
- Convert PAN predefined applications references to Versa predefined applications
- Convert PAN custom URL categories objects to Versa format
- Convert PAN URLF profiles configuration to Versa format
- Recognize if policies are using security profiles and configure the Versa equivalent

## Before you execute the conversion:

When you untar the file, there is a sample PAN configuration that you can use to run the conversion. The sample configuration file gives you an idea of the configuration type that the scripts can work with.

## Pre-requisites:

Before you execute the conversion process, these are the pre-requisites that the scripts need during conversion process:

- Export PAN configuration in text file. Make sure you export not only the firewall policies, but also all dependent objects. Eg, ip-objects, profiles, groups etc... Aggregate everything in a single file named "source-pan-rules.txt", the scripts are designed to correctly recognize the different

configuration and extract what they need for conversion. From version 1.2, you can copy the PAN file "sp-config.xml" to the main directory and the scripts will automatically convert it

- If the policies are using LDAP users/groups, you must export all LDAP users and groups from Active Directory or from VOS. The objects must be in DN format. How you package the file doesn't matter, the scripts are designed to recognize and extract. The only requirement is that you group DN objects into 2 separate sections. One section for users and the other for groups. The section must be separated by these delimiters (without the quotes): "----BEGIN USERS----", "----END USERS----", "----BEGIN GROUPS----" and "----END GROUPS----". Save the file as "ldap-source-input.txt. There's a sample LDAP file for your reference. You can extract the LDAP users and groups on the VOS with these commands: `show orgs org-services <your-org> user-identification ldap-server-data users brief | nomore` and `show orgs org-services <your-org> user-identification ldap-server-data groups brief | nomore`"
- Review the file: "../miscellaneous/PAN-to-Versa-applications.txt". This file list some often-used predefined applications that be used by the scripts to convert PAN predefined applications to Versa. Add more applications in this file if needed.
- Review the file: "../miscellaneous/PAN-to-Versa-urll-categories.txt". This file list all the current Versa URL categories as of writing of this document. It's a one-one conversion from PAN to Versa
- Review the file: "../miscellaneous/PAN-to-Versa-services.txt". This file list some often-used predefined services that be used by the scripts to convert PAN predefined applications to Versa. Add more services in this file if needed.
- The scripts must be run on MacOS or Linux with Python v3.9 or later.

#### Instruction:

To start the conversion process, run the wrapper script `./RunMe.sh`. The wrapper script will automatically run all necessary scripts located in sub-directory `./scripts`. As stated before, the whole process is automated, except for when there are unresolvable elements that need your intervention. Those exceptions will be described below.

The whole conversion process framework is designed to be modular and divided into many steps. Some steps will require your input and some steps will output information for your review.

#### Description:

- **Step0:** this step will clean the input file "source-pan-rules.txt" for any undesirable characters that will affect the conversion process later. It will list all firewall policies that use security profiles, this is informational. The script will convert these security profiles configuration to Versa default. Lastly, it will output the file "zone-conversion.txt". YOU MUST manually type in the corresponding zones on Versa platform.
- **Step1:** this step will clean and convert all the policies names to Versa supported format. It will prompt you for the maximum length that you wish to use, default is 63 characters. The script will truncate any policy name to that length. It will also clean and replace any unsupported characters with `"_"`. If you wish to review which policies were affected, there's an output file `../step-1/ step-1_modified-policy-name.txt`
- **Step2:** this step mainly focuses on ip-objects and groups of objects extraction and sanitizing. It will extract all those objects, truncate the objects names to maximum 63 characters, replace any unsupported characters with `"_"`. It will automatically add `/32` to any ip-address object that doesn't have the `/32` subnet mask. Lastly, it will output 2 files that list all objects that the script

can't resolve, "must-fix-address-object.txt" and "must-fix-address-group-object". YOU MUST review and resolve those objects later on before the conversion.

- **Step3:** this step will correlate all ip-address objects and objects groups with the actual firewall policies. It will remove all objects that are not used by the policies or any ip-address objects that are not used in any objects groups. The script is able to recognize and correlate nested groups, that is groups within groups. Note: this step could take a while if you have thousands of objects.
- **Step4:** this step mainly focuses on custom service objects. It will extract and sanitize the objects names. It will truncate any object name that's longer than 31 characters. It will replace all unsupported characters with "\_". It will correlate with the actual firewall policies and change the reference name in the policies. The script will output any service object configuration that the script can't resolve or recognize. Review the file "../step-4/ step-4\_unsupported-service-config.txt"
- **Step5:** since Versa do not support services groups, the script will extrapolate the group service members and replace all firewall policies reference with the corresponding individual service object names. This step will also extract predefined service and application objects and it will convert PAN predefined service and application to Versa's version based on the files "../miscellaneous/ PAN-to-Versa-applications.txt" and "../miscellaneous/ PAN-to-Versa-services.txt". If there are no equivalent service or application in the conversion file, the script will use whatever object name that's currently in the PAN configuration. **Note: those files contain some of the mostly used services and applications, it's not complete. YOU MUST, edit the files and insert the Versa equivalent. Else, the import of the service template will fail if you're not using the correct service or application name.**
- **Step6:** this step will extract custom URL categories configuration. It will sanitize the object names, truncate the length and replace unsupported characters with "\_". The script will also rectify any modified names in the firewall policies reference.
- **Step7:** this step will extract all LDAP users and groups from the firewall policies and it will divide the LDAP objects into separate users and groups based on the reference file "ldap-source-input.txt". The script is able to recognize LDAP DN, UPN and canonical formats and it will process the objects accordingly.
- **Step8:** this step will sanitize and convert the LDAP configuration lines to Versa format. It will also convert any non-DN users/groups format in the policies to DN format.
- **Step9:** this step will convert and transfer all PAN address objects to Versa service template. It will also do a final cleanup and CIDR formatting of any objects with the wrong subnet mask.
- **Step10:** this step will convert and transfer all PAN groups objects to Versa service template.
- **Step11:** this step will convert and transfer all PAN custom services objects to Versa service template.
- **Step12:** this step will convert and transfer all PAN custom URL categories objects to Versa service template.
- **Step13:** this step will convert and transfer all PAN security URLF profiles to Versa service template. This will also output any unsupported predefined URL categories in the file "unknown-urlf-category.txt"
- **Step14:** this is the last step where the script will transfer and convert PAN firewall policies to Versa template.